



Innovation Project

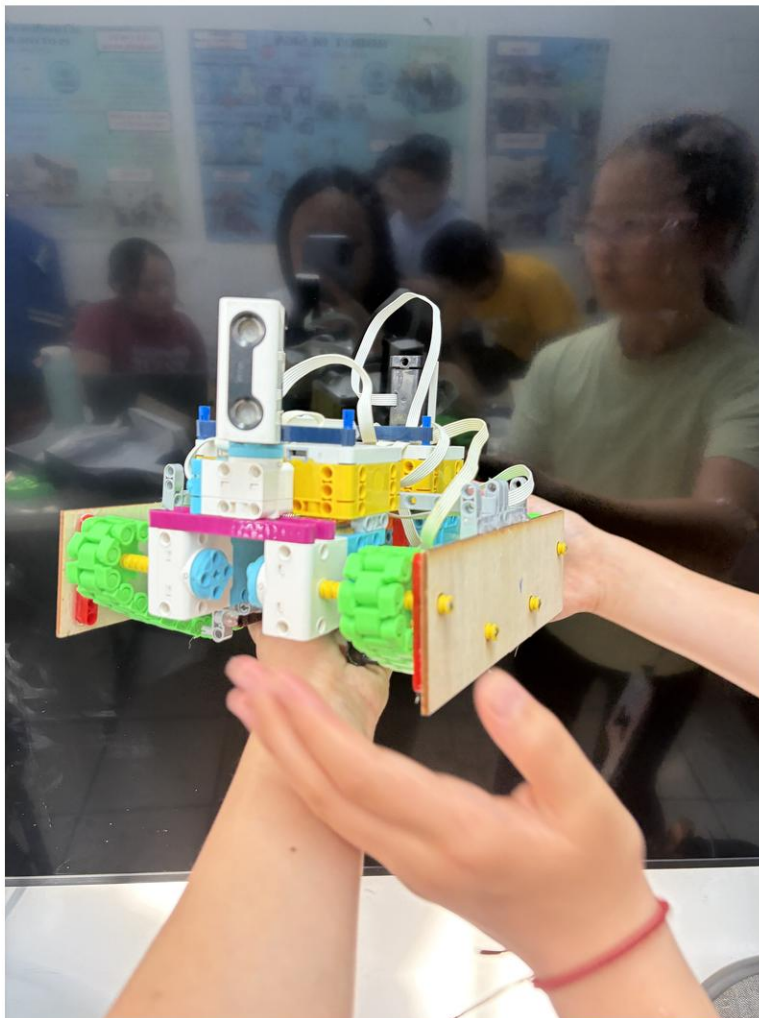
Safer Mapping for Fragile Archaeological Sites

Rover design, camera mapping, expert feedback, and future VR

JUDGE QUICK SUMMARY

What problem did we solve?

Archaeologists need safer ways to document dark, unstable, tight spaces without putting people or fragile cultural heritage at risk.



- Solution: remote rover concept for observation and mapping.
- Current upgrade: recent wood-panel rover platform with sensor mast.
- Programming upgrade: camera-first mapping instead of distance-sensor-first mapping.
- Impact: safer exploration and future VR walkthroughs.

IDENTIFY

Real archaeology risks

We researched artifact fragility, difficult field conditions, dark caves, and expert concerns before choosing our problem.

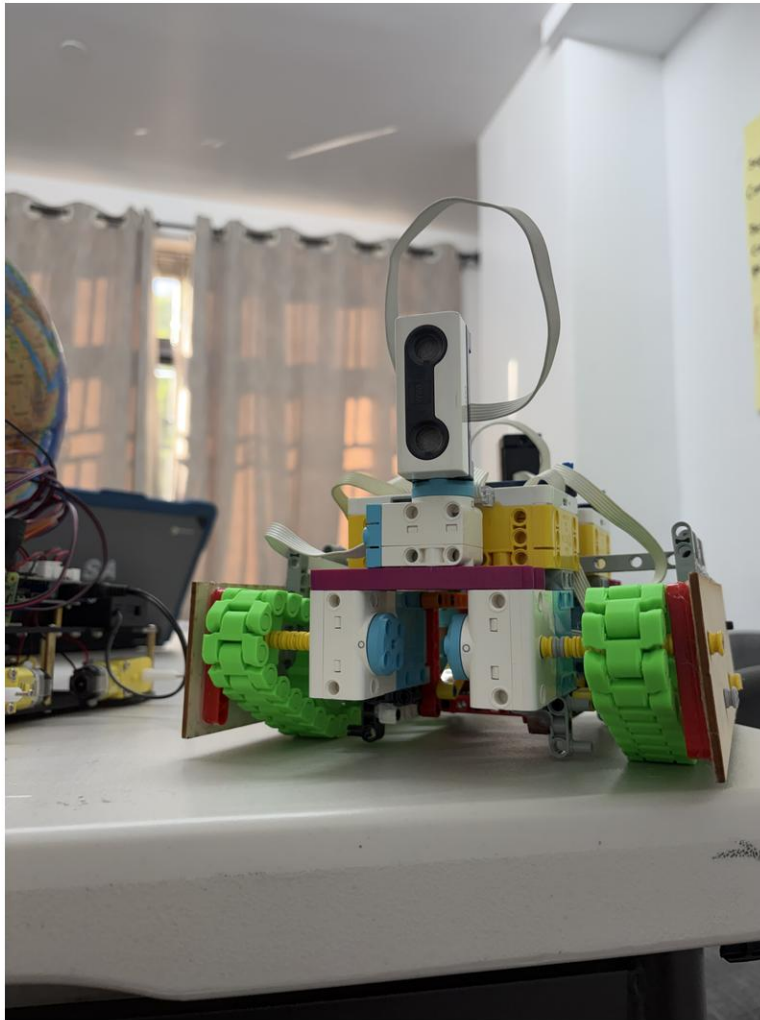


- Museum visit helped us understand preservation.
- Expert feedback shaped camera placement and modularity.

SOLUTION

A rover for safer mapping

The rover sends cameras and mapping tools into spaces people should not enter first.



- Stable movement.
- Camera and mapping focus.
- Repairable modular platform.

PROTOTYPE EVOLUTION

Physical + digital versions

Each prototype answered a different engineering question.



- Cardboard: quick layout.
- LEGO: movement test.
- Wood-panel rover: stronger current platform.
- Electronics rover: sensor experiments.

LATEST UPGRADE

Recent rover + camera mapping

The current workflow combines the recent rover build with camera-based mapping output.



- Recent table demo version.
- OpenCV stitched room model.
- Camera logic gives richer context than distance-only readings.

ACTUAL STEM LAB SCAN

Our real room panorama

This is the actual STEM lab panorama produced by the guided camera mapping prototype.



- Captured from the team workspace.
- Stitched from overlapping camera frames.
- Shows the mapping workflow working on a real indoor space.

IMPACT

Protect people and history

The project reduces risk while helping archaeologists and museums document and share fragile places.



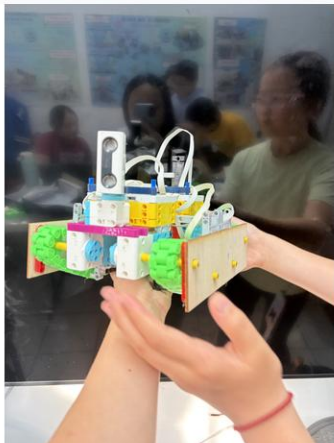
- Safer pre-entry documentation.
- Better visual context.
- Future VR education experience.

CREATE

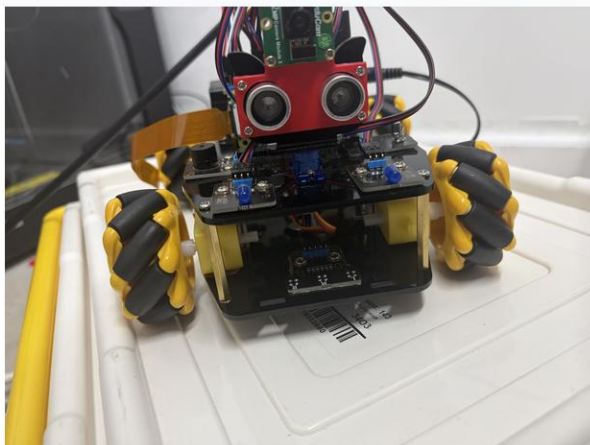
Prototype evidence



Recent wood-panel rover



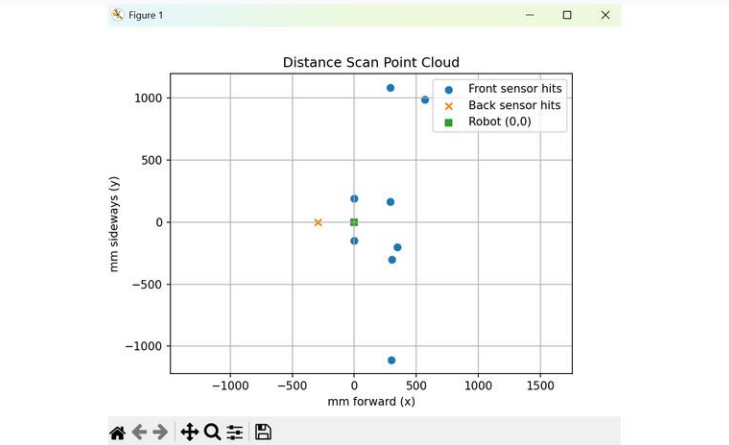
Team-held current prototype



Electronics rover front view

PROGRAMMING

Mapping evidence



Early distance baseline



Captured camera frames



Stitched panorama output

CURRENT MAPPING SYSTEM

How our tunnel scan works now

The current prototype guides overlapping camera captures, then stitches the saved frames into a wider visual map.

Logic summary	
Open camera	OpenCV reads 640 x 480 frames at 15 FPS.
Detect motion	ORB feature points compare the previous capture to the current frame.
Guide capture	Shift ratio and match count decide turn right, move slower, or READY.
Stitch output	OpenCV Stitcher creates the panorama and a contact sheet.

Code / math view

```
MIN_SHIFT_RATIO = 0.14
MAX_SHIFT_RATIO = 0.42
MIN_GOOD_MATCHES = 25

median_shift = median(previous_x - current_x)
shift_ratio = median_shift / frame_width
```

Good tunnel scans need enough shared visual features and the right amount of overlap.

MAPPING MATH

From readings to map points

The early distance-sensor version helped us explain the math of turning tunnel measurements into plotted points.

Logic summary	
Angle	$\text{angle_deg} = \text{angle_index} \times 15$
Front point	$x = r \cos(\text{theta}), y = r \sin(\text{theta})$
Back point	$\text{theta_back} = \text{theta} + 180$
No hit	Ignore readings ≥ 2000 mm.

Code / math view

```
a_deg = angle_index * ANGLE_STEP_DEG
th = radians(a_deg)
x = distance_mm * cos(th)
y = distance_mm * sin(th)

# back sensor uses the opposite direction
th_back = radians(a_deg + 180)
```

This converts rover-relative measurements into a local 2D point cloud.

What the next system would look like

The future mapper would combine images, pose estimates, obstacle marks, and VR export.

Logic summary	
Keyframes	Save useful views only when overlap and image quality are good.
Pose estimate	Track turn angle and distance traveled between frames.
Obstacle map	Mark open and blocked areas as the rover advances.
VR export	Turn the scan into an explorable tunnel walkthrough.

Code / math view

```
for each camera frame:
  features = detect_features(frame)
  pose = estimate_motion(previous_frame, frame)
  if overlap_is_good(pose, features):
    save_keyframe(frame, pose)
    update_tunnel_map(keyframes)
    mark_open_space_and_obstacles()
  export_panorama_and_vr_scene()
```

This is the roadmap from camera demo to a real tunnel mapping system.

JUDGE REFERENCE

Questions this book helps answer

- What real-world problem did you identify?
- How did expert feedback change your design?
- What did each prototype teach the team?
- Why did you move from distance sensors to camera mapping?
- How could this help archaeologists, museums, or students?